C1 past final worksheet

1. (Fall 2019, #4) Consider $f(x) = \dfrac{2}{3-x}$ with domain $[0,2]$. Note that, the range of f is contained in $[0,2]$. (You do not need to show this fact and may use it freely.) Let $u_0, u_1, u_2, \ldots$ be the sequence defined by $u_0 = \dfrac{3}{2}$ and $u_{n+1} = f(u_n)$, for all $n \geq 0$.

   (a) Show that, for any $n \geq 0$, we have $0 \leq u_n \leq 2$. You **must** use a mathematical induction to prove this assertion.

   (b) Show that the sequence $u_0, u_1, u_2, \ldots$ is decreasing. You **must** use mathematical induction to prove this assertion. **Hint:** Use the fact that f is increasing.

   (c) Show that the sequence $u_0, u_1, u_2, \ldots$ converges to a limit l, then determine the value of l. You must carefully justify your answer.

2. (Fall 2019, #5c) Let $0 < r < s$ be constants. Does the sequence $\{(r^n + s^n)^{1/n}\}$ converge as $n \to \infty$? If it does converge, what is the limit?

3. (Spring 2013, #5)

(a) If $a_n = \dfrac{\sqrt{n}\sin(n!)}{n+1}$, carefully determine if $\lim_{n\to\infty} a_n$ exists, and if so, find it.

(b) Let $c > 0$, set $b_1 = c$, $b_2 = \sqrt{c} + 1$ and in general let $b_{n+1} = \sqrt{b_n} + 1$.

i. Show that the sequence (b_n) is increasing or decreasing, depending on c.

ii. Show that (b_n) is bounded.

iii. Justify whether (b_n) converges or diverges.

iv. If (b_n) converges find its limit.

4. (Spring 2011, #6) In each case below, the n=th term of a sequence is given. Give reasons why the sequence diverges, or why it converges and find its limit.

   (a) $a_n = \left(\dfrac{4n-3}{n} - \dfrac{3n}{n+1} \right)^{n^2+n}$

   (b) $b_n = (3 + \sin(n))^{1/n}$

   (c) $c_n = (-1)^n(1 - (1/n))$

5. (Spring 2010, #6) Determine whether each of the following sequences is convergent or divergent. Find the limit if the sequence is convergent. (Remember: **You must show all your work and justify your answers to get full credit)**

(a) $a_n = \dfrac{(3\cos(n))^n}{5^n + 2^n}$

(b) $a_n = (-1)^{n+1}\dfrac{4n^2+1}{5n-n^2}$